Singly linked node

A singly linked node stores data and a next pointer. Starting at head, follow next to visit subsequent nodes. In a non-circular singly linked list, the last node has next equal to NULL.

Doubly linked node

A doubly linked node stores data, prev, and next. Prev refers to its predecessor, while next refers to its successor. A prev link allows backward traversal without restarting from head.

Insertion at the start

To insert at the start of a singly linked list, set newNode->next = head and then head = newNode. This also works when the old head is NULL. No traversal to the last node is necessary.

Insertion at the end

For insertion at the end of a singly linked list, first set newNode->next to NULL. If head is NULL, assign head = newNode. Otherwise traverse until temp->next is NULL, then set temp->next = newNode. The existing head does not change in the nonempty case.

Circular traversal

In a circular singly linked list, the last node links back to head. A complete traversal stops after returning to its starting node. Waiting for NULL would fail to terminate in a correctly formed nonempty ring.